perplexity

Plan: Suppliers + Simple Stock Receipt

Build this as one Inventory vertical slice: **manage suppliers, receive stock from a supplier into a branch, and automatically create an auditable stock movement.** Do not build purchase orders, supplier invoices, approvals, or supplier performance reports in this first release.

The goal is simple: a merchant should be able to say, "I received 20 bags of rice from Supplier X into my Ikeja store," and Orvio should increase stock accurately while preserving who received it, when, from where, and at what cost.

1. Product outcome

After this feature, an authorized Inventory user can:

- Add and manage suppliers for their organization.
- Record a stock receipt from a supplier into a specific branch.
- Add one or more products to the receipt.
- Enter received quantity and optional unit cost.
- See the branch stock increase immediately.
- View the receipt in stock history, linked to the supplier and receiving user.
- Search and filter receipts by supplier, branch, product, date, and status.

This keeps replenishment simple for small merchants while creating the data foundation for purchase orders and GRNs later. A stock receipt should record supplier, SKU, received quantity, condition, receipt date, and receiving user; accepted goods are what increase stock. [1] [2]

2. MVP scope

Must have

- Supplier create, view, edit, deactivate, and search.
- Supplier list with name, phone, email, status, and product/receipt count where available.
- Simple stock receipt creation.
- Multi-product receipt: add several products to one receipt.
- Received quantity per product.
- Optional unit cost per product.
- Optional supplier invoice/receipt number and notes.
- Automatic stock increase at the selected branch.

- Stock movement records linked to the receipt.
- Receipt history and detail view.
- Organization and branch permission enforcement.
- Audit metadata: created by, received by, timestamp, organization, branch, supplier.

Should have

- Receipt reference auto-generation, such as `RCPT-000123`.
- CSV export of receipts.
- Low-stock indicator visible after receipt.
- Duplicate-supplier warning based on name or phone.
- Receipt status: `completed` only for MVP; draft status can be deferred.

Not in this release

- Purchase orders.
- Goods Received Notes linked to POs.
- Supplier invoices and payment tracking.
- Partial delivery, backorder, or over-delivery reconciliation.
- Supplier returns.
- Batch, expiry, serial, and lot tracking.
- Supplier ratings and performance analytics.
- Multi-currency supplier pricing.
- Approval workflow for receipts.

3. User roles and permissions

Action	Owner	Inventory Manager	Storekeeper	Cashier	Viewer
View suppliers	Yes	Yes	Yes	No	Yes
Create/edit supplier	Yes	Yes	No	No	No
Deactivate supplier	Yes	Yes	No	No	No
Create stock receipt	Yes	Yes	Yes	No	No
View receipt history	Yes	Yes	Yes	Limited	Yes
Export receipts	Yes	Yes	No	No	Yes
Delete posted receipt	No	No	No	No	No

Rules:

- The organization owner has access to all branches.

- Manager and Storekeeper access must be limited to branches assigned to them.
- A Cashier should not receive stock or view supplier cost information.
- A Viewer can see receipt history but cannot create or edit records.
- A posted receipt cannot be deleted; corrections should create a new adjustment or reversal movement.

4. User journeys

Journey A — Add a supplier

1. Owner or Manager opens **Inventory → Suppliers**.
2. Clicks **Add supplier**.
3. Enters supplier name, phone, email, address, and optional contact person.
4. Saves the supplier.
5. Supplier appears in the supplier list with status **Active**.

Journey B — Receive stock

1. Storekeeper opens **Inventory → Stock → Receive stock**.
2. Selects an authorized branch.
3. Selects a supplier.
4. Adds one or more products.
5. Enters received quantity and optional unit cost for each product.
6. Adds optional invoice number and note.
7. Reviews the receipt summary and confirms.
8. System creates the receipt, increases branch stock, and creates stock movements.
9. User sees a success screen with the receipt reference and a link to view stock.

Journey C — View receipt history

1. User opens **Inventory → Stock → Receipts**.
2. Filters by branch, supplier, date, product, or status.
3. Opens a receipt.
4. Sees supplier, branch, received-by user, products, quantities, unit costs, notes, and linked stock movements.

5. Functional requirements

FR-1 — Supplier management

- Users with permission can create a supplier.
- Supplier name is required.
- Phone or email should be required; confirm the final rule before implementation.
- Supplier records belong to the organization.
- Supplier name should be unique within the organization, or the system should warn on a likely duplicate.
- Suppliers can be deactivated but not hard-deleted if they have receipt history.
- Deactivated suppliers cannot be selected for new receipts.
- Existing receipts must continue to display the supplier details.

FR-2 — Create simple stock receipt

- User selects an active branch they are authorized to access.
- User selects an active supplier.
- User adds one or more active products.
- Each line requires a positive received quantity.
- Unit cost is optional in the first release; if entered, it must be zero or greater.
- Receipt date defaults to now but can be edited within a defined allowed period.
- The user can add an invoice number and note.
- The system prevents submitting a receipt with zero lines.
- On confirmation, the receipt is created as `completed`.

FR-3 — Update stock balances

- For each accepted receipt line, increase the product's quantity on hand at the selected branch.
- Stock balance and movement must be written in one transaction.
- The system must not create a receipt without corresponding stock movements.
- If any line fails validation, no part of the receipt should be committed.
- The system must handle concurrent receipts safely so two users receiving stock do not corrupt the balance.

FR-4 — Stock movement traceability

Each receipt line creates a movement containing:

- Organization ID.
- Branch ID.
- Product ID.
- Movement type: `in` or `receive`, aligned with the existing stock-movement model.
- Positive quantity.
- Supplier reference.
- Receipt reference.
- Received-by user.
- Timestamp.
- Optional note.

The movement should never be edited or deleted after posting. Corrections happen through a separate stock adjustment or reversal record.

FR-5 — Receipt history

- Users can view receipts for branches they can access.
- Users can filter by date range, branch, supplier, product, and status.
- Users can search by receipt reference, supplier name, product name, or SKU.
- Receipt list uses server-side pagination.
- Receipt detail shows all lines, quantities, costs where authorized, notes, and linked movements.

FR-6 — Supplier receipt visibility

- Supplier detail page shows receipt history for that supplier.
- Product stock history shows the supplier on inbound receipt movements.
- Branch stock history shows the receipt reference and supplier.

6. Data model

```
suppliers
  id
  organization_id
  name
  phone
  email
  address
  contact_person
  tax_id
  notes
```

```
  status
  created_by
  created_at
  updated_at
  deleted_at

stock_receipts
  id
  organization_id
  branch_id
  supplier_id
  reference
  receipt_date
  supplier_invoice_number
  notes
  status
  created_by
  received_by
  created_at
  updated_at

stock_receipt_items
  id
  stock_receipt_id
  product_id
  quantity_received
  unit_cost
  notes
  created_at

stock_movements
  id
  organization_id
  branch_id
  product_id
  stock_receipt_id
  movement_type
  quantity
  reason
  reference_type
  reference_id
  created_by
  created_at
```

Use the existing `stock_items` table for the product-branch balance. The receipt should update `quantity_on_hand` and create the corresponding movement in the same transaction.

7. API plan

Use Orvio's existing REST, `/api/v1`, authentication, Zod validation, response-envelope, and cursor-pagination conventions. The endpoint names below are conceptual and must be reconciled with the current API contract.

Capability	Conceptual endpoint	Notes
List suppliers	`GET /suppliers`	Filter by search, status; cursor pagination
Create supplier	`POST /suppliers`	Owner/Manager only
Get supplier	`GET /suppliers/:id`	Include receipt history link
Update supplier	`PATCH /suppliers/:id`	Owner/Manager only
Deactivate supplier	`POST /suppliers/:id/deactivate`	Soft delete/deactivate
List receipts	`GET /stock/receipts`	Branch/supplier/product/date filters
Create receipt	`POST /stock/receipts`	Atomic receipt + stock update + movements
Get receipt	`GET /stock/receipts/:id`	Include lines and movement references

8. Validation and error handling

- Reject a receipt if the user lacks branch access.
- Reject a receipt if the supplier belongs to another organization.
- Reject a receipt if the branch or product belongs to another organization.
- Reject zero, negative, or non-numeric quantities.
- Reject a receipt with no product lines.
- Reject a receipt using an inactive supplier or inactive product.
- Reject invalid unit cost values.
- If stock update fails, roll back the receipt and all movements.
- Show clear field-level validation errors.
- Show a retryable error for temporary API/database failure, without creating a partial receipt.

9. Security and audit

- Derive organization ID and user ID from the authenticated session; never trust client-supplied identity fields.
- Enforce active organization membership and Inventory branch assignment before allowing receipt creation.
- Enforce role permissions server-side.
- Scope every supplier, receipt, product, branch, and movement query by `organization_id`.
- Test cross-organization access and cross-branch access.
- Record the actor and request correlation ID for receipt creation.
- Do not log secrets or expose internal stack traces.
- Preserve receipt and movement history for audit and reconciliation.

10. UI requirements

Suppliers screen

- Search bar.
- Supplier table: name, phone, email, status, receipts count, created date.
- Add Supplier button.
- Supplier detail drawer or page.
- Empty state: “Add your first supplier to start tracking where stock comes from.”

Receive Stock screen

- Branch selector.
- Supplier selector.
- Product search/add rows.
- Quantity and optional unit-cost fields.
- Invoice number and notes.
- Receipt summary: total lines, total quantity, total cost if costs are entered.
- Confirm button with loading state.
- Success state with receipt reference and “View stock” action.

Receipts screen

- Filters: branch, supplier, date range, product, status.
- Receipt table: reference, date, supplier, branch, product count, total quantity, received by.
- Receipt detail view.
- Empty state: “No stock receipts yet. Receive your first stock delivery.”

11. Acceptance criteria

- An authorized Owner, Manager, or Storekeeper can create a supplier.
- A supplier cannot be created or accessed outside its organization.
- An authorized user can create a receipt for a branch they can access.
- A receipt with multiple products increases stock for every valid product at the selected branch.
- Stock balance, receipt, receipt items, and stock movements are committed atomically.
- A receipt cannot be submitted with zero, negative, or invalid quantities.
- A Cashier cannot create a receipt or access supplier cost data.
- A Storekeeper assigned to Branch A cannot receive stock into Branch B.

- A receipt appears in receipt history, supplier history, product stock history, and branch stock history.
- A deactivated supplier cannot be selected for a new receipt but remains visible on historical receipts.
- Refreshing, retrying, or navigating away does not duplicate stock movement.
- Concurrent receipts update stock correctly without negative balances or lost updates.

12. Test plan

Unit tests

- Supplier validation.
- Receipt line validation.
- Quantity and cost validation.
- Stock-balance calculation.
- Receipt reference generation.
- Permission mapping.

API/integration tests

- Create supplier.
- Update and deactivate supplier.
- Create receipt with one product.
- Create receipt with multiple products.
- Reject invalid supplier, branch, product, quantity, or permission.
- Confirm stock balance increases correctly.
- Confirm movement records are created.
- Confirm transaction rollback when a line fails.
- Confirm concurrent receipts do not corrupt balances.
- Confirm cross-organization and cross-branch denial.

End-to-end tests

- Owner creates supplier.
- Storekeeper receives stock into assigned branch.
- Stock dashboard/product list reflects the new quantity.
- Receipt appears in history.
- Unauthorized user cannot receive stock or view restricted supplier data.

13. Delivery milestones

Milestone	Scope	Exit criteria
M1 — Supplier foundation	Supplier schema, CRUD API, supplier UI, permissions	Authorized users can manage suppliers; cross-org access denied
M2 — Receipt data model	Receipt, receipt item, movement linkage, API contract	Schema and API contract reviewed
M3 — Receive stock flow	Receive Stock UI, validation, atomic stock update	One and multi-product receipts work end to end
M4 — History and visibility	Receipt list/detail, supplier/product/branch history links	Receipts traceable across all relevant views
M5 — Hardening	Concurrency, idempotency, RBAC, tenant isolation, E2E tests	No partial receipts, duplicate movements, or unauthorized access

14. Open decisions

Confirm these before implementation:

1. **Supplier uniqueness:** Should supplier name be strictly unique per organization, or should the system only warn about possible duplicates?
   Recommendation: Warn on duplicate name/phone, but allow save.
2. **Required supplier contact:** Must a supplier have phone, email, or both?
   Recommendation: Require name; phone or email is strongly recommended but optional for MVP.
3. **Unit cost:** Should unit cost be optional or required?
   Recommendation: Optional for MVP, because many small merchants may not know cost at receipt time.
4. **Receipt editing:** Can a user edit a completed receipt?
   Recommendation: No. Use a reversal/adjustment flow instead.
5. **Receipt date:** Can users backdate receipts?
   Recommendation: Allow limited backdating, such as within 30 days, but confirm the business rule.
6. **Supplier deletion:** Should deletion be allowed at all, or only deactivation?
   Recommendation: Deactivation only once receipts exist.
7. **Cost visibility:** Should Storekeeper see unit cost?
   Recommendation: Yes for receiving, but hide aggregate profit/margin reporting unless the role policy allows it.
8. **Receipt numbering:** Use auto-generated reference such as `RCPT-000123`, or allow user-entered reference?
   Recommendation: Auto-generate; allow optional supplier invoice number separately.

15. Definition of done

The feature is complete when:

- Supplier CRUD works with correct organization scoping and permissions.
- A user can receive stock from a supplier into an authorized branch.
- Stock balance and movement are updated atomically.
- Receipts are searchable, filterable, and viewable in history.
- Supplier, product, branch, and receipt records are properly linked.
- Unauthorized users cannot create receipts or access restricted data.
- Cross-organization and cross-branch access tests pass.
- Concurrent receipt tests pass.
- Loading, empty, success, validation, and error states are implemented.
- Unit, API/integration, and E2E tests pass.
- The implementation follows Orvio's existing API, database, RBAC, audit, and modular-monolith conventions. [3] [4] [5]

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